

LAKEVILLE AIRLAKE, MN, USA

WMO: 726562

Lat: 44.628N

Lon: 93.228W

Elev: 293

StdP: 97.86

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04943

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-22.6	-29.1	0.3	-24.3	-27.0	0.3	-22.5	11.6	-9.8	10.4	-8.5	2.5	280	0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.2	23.7	30.3	22.6	28.7	21.9	25.5	29.8	24.2	28.3	23.2	27.0	4.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.6	28.1	22.8	18.1	26.6	22.2	17.5	26.1	79.9	30.5	74.6	28.3	70.3	27.0	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.5	7.6	DB	-28.8	34.9	4.0	1.8	-31.7	36.2	-34.1	37.3	-36.4	38.3	-39.3	39.6	(4)
(5)			WB	-29.0	27.4	3.9	1.7	-31.8	28.7	-34.1	29.7	-36.3	30.6	-39.2	31.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-9.6	-7.4	-0.4	7.3	14.3	20.0	22.0	20.4	16.4	9.1	1.2	-6.3	(6)	
	DBStd	12.37	7.34	7.32	6.95	5.64	4.85	3.67	3.17	3.16	4.64	5.19	5.93	6.85	(7)	
	HDD10.0	2409	607	487	328	114	14	0	0	0	5	81	270	504	(8)	
	HDD18.3	4343	865	720	580	331	144	23	6	16	89	292	514	762	(9)	
	CDD10.0	1437	0	0	6	34	147	299	373	322	198	52	5	0	(10)	
	CDD18.3	328	0	0	0	2	18	73	121	79	32	4	0	0	(11)	
	CDH23.3	3000	0	0	1	26	221	658	1108	641	302	43	1	0	(12)	
	CDH26.7	903	0	0	0	6	70	208	360	168	82	9	0	0	(13)	
(14) Wind	WSAvg	3.1	3.4	3.5	3.5	3.9	3.5	2.8	2.2	2.0	2.5	3.1	3.3	3.2	(14)	
(15) Precipitation	PrecAvg	733	21	18	45	64	88	104	93	95	79	56	42	24	(15)	
	PrecMax	1044	70	64	105	158	158	240	294	174	206	134	134	112	(16)	
	PrecMin	357	2	0	8	7	17	8	26	17	8	6	1	2	(17)	
	PrecStd	158	17	15	26	33	34	59	53	42	49	38	33	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	11.2	21.1	27.1	32.3	33.1	34.0	32.6	32.1	27.8	19.7	11.0	(19)	
		MCWB	3.8	8.2	15.0	15.9	20.3	23.1	26.3	24.3	22.3	18.4	12.7	8.1	(20)	
	2%	DB	3.0	7.2	16.2	22.5	28.0	31.3	32.3	30.2	28.9	23.4	16.0	7.2	(21)	
		MCWB	1.2	4.7	12.0	12.9	18.1	22.2	25.3	23.5	21.5	16.6	11.0	4.9	(22)	
	5%	DB	2.1	4.0	12.6	19.1	26.0	28.9	30.2	28.0	27.2	20.9	12.6	4.0	(23)	
		MCWB	0.9	2.4	9.5	11.6	16.8	20.9	23.7	22.2	20.0	14.7	8.9	2.2	(24)	
	10%	DB	0.2	2.6	8.9	17.1	22.7	27.3	28.6	27.2	24.5	17.6	9.1	2.5	(25)	
		MCWB	-0.4	1.3	6.5	10.4	15.2	19.5	22.9	21.5	18.8	13.0	6.5	1.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	8.0	16.1	16.8	22.6	25.5	28.2	26.5	24.3	20.9	14.9	9.0	(27)	
		MCDB	6.6	9.8	19.1	24.4	29.2	31.1	32.2	30.0	29.0	25.0	18.8	10.4	(28)	
	2%	WB	1.9	4.8	12.9	14.8	20.1	23.8	26.4	24.9	22.8	17.9	11.9	4.7	(29)	
		MCDB	3.1	6.7	15.4	20.4	25.6	28.6	30.6	28.5	26.7	21.2	15.0	6.7	(30)	
	5%	WB	0.8	2.9	9.7	12.8	18.3	22.3	24.8	23.6	21.5	15.6	9.1	2.8	(31)	
		MCDB	1.8	4.1	12.8	17.6	22.8	27.3	28.9	27.1	25.3	19.5	12.3	4.1	(32)	
	10%	WB	-0.6	1.3	6.5	11.1	16.8	20.9	23.6	22.5	19.9	13.4	7.1	1.4	(33)	
		MCDB	0.2	2.6	9.0	15.7	21.3	25.3	27.4	25.7	23.8	17.3	9.1	2.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.4	9.4	11.0	11.3	10.6	10.9	11.0	12.0	10.7	8.9	7.9	(35)	
		MCDBR	9.3	9.6	13.4	16.1	15.5	13.0	12.2	12.2	14.1	15.5	13.4	9.5	(36)	
	5% WB	MCWBR	8.0	7.8	9.1	8.7	7.2	6.1	6.5	6.7	7.4	8.7	9.0	7.9	(37)	
		MCWBR	8.8	8.8	12.6	13.8	12.5	11.3	11.1	10.9	11.9	12.6	12.5	8.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.270	0.288	0.321	0.376	0.404	0.403	0.404	0.404	0.377	0.323	0.293	0.269	(40)	
		taud	2.420	2.351	2.383	2.290	2.267	2.313	2.323	2.322	2.363	2.508	2.502	2.439	(41)	
	Edn at Noon	Edn at Noon	861	903	915	885	867	867	861	847	842	855	828	824	(42)	
		Edh at Noon	72	93	104	124	132	126	124	119	106	80	67	64	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.74	2.82	3.73	4.57	5.32	5.95	6.32	5.28	4.14	2.65	1.69	1.23	(44)	
	RadStd	RadStd	0.21	0.30	0.36	0.48	0.43	0.48	0.24	0.37	0.30	0.35	0.17	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page